

Information Visualization and Visual Analytics (M1522.000500)

Multidimensional Data Visualization

Jinwook Seo, Ph. D.

Professor, Dept. of Computer Science and Engineering
Seoul National University

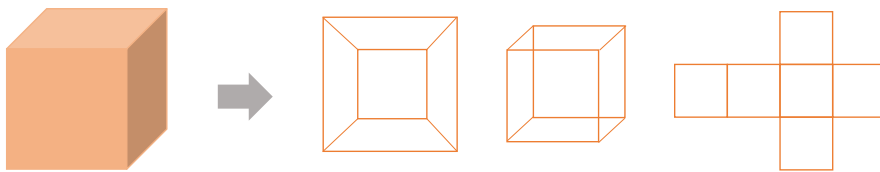
Multidimensional Data Visualizations

Multidimensional Data Visualization

- Number of variables per data case (item)
 - 1 – Univariate data
 - 2 – Bivariate data
 - 3 – Trivariate data
 - > 3 – Multivariate (Hypervariate) data
- General representation techniques for multivariate (> 3 variables) per data case

Displaying Multidimensional Data in 2D Space

- Fundamentally, we have 2 geometric (position) display dimensions
- For data sets with > 2 variables, we must project data down to 2D
- Need visual mapping that locates each dimension into 2D plane
- Computer Graphics: 3D $\rightarrow$ 2D representation



Even a 3D cube cannot be exactly projected in 2D space!!

Approaches

- A spreadsheet already does that too
 - Each variable is positioned into a column
 - Data cases in rows
 - This is a projection (mapping)
- What else?

Multiple Views

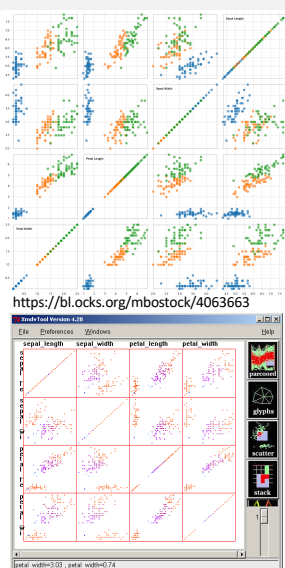
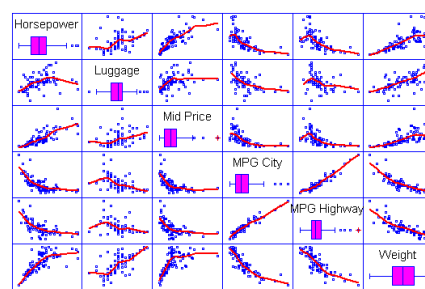
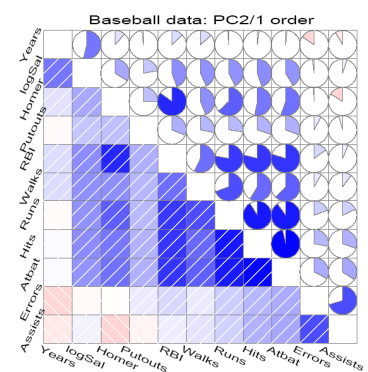


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Multidimensional Data Visualizations

Scatterplot Matrix (SPLOM)


<https://bl.ocks.org/mbostock/4063663>

<http://statvision.com/eda.htm>


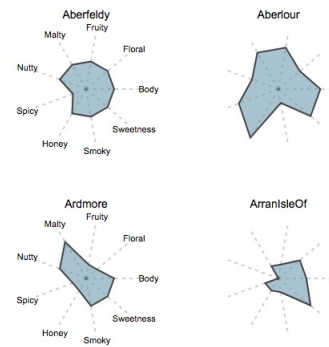
Correlogram for Baseball data
(from <https://www.datavis.ca/papers/corrogram.pdf>)

Scatterplot Matrix in XmdvTool

Information Visualization and Visual Analytics – Multidimensional data Visualization

Glyphs

- “a graphical object designed to convey multiple data values”
- “composite graphical objects where different geometric and visual attributes are used to **encode** multidimensional data structures in combination”
- *Chernoff Faces*: most prominent example
- Arrows → primitive example
 - length, width, angle, and color



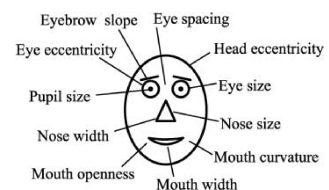
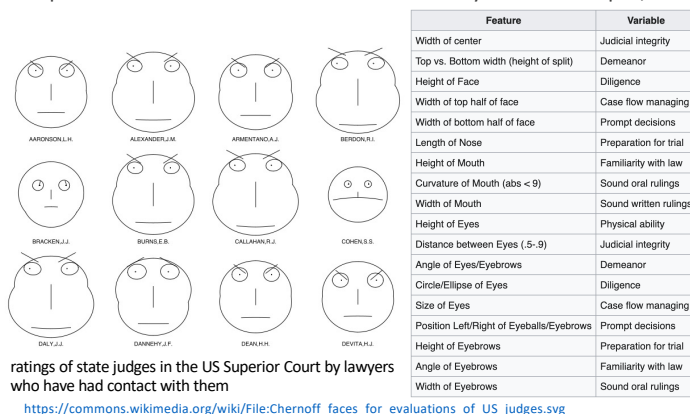
Herman Chernoff (1973). "The Use of Faces to Represent Points in K-Dimensional Space Graphically". Journal of the American Statistical Association 68 (342): 361–368

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Glyphs

Chernoff Faces

- Encode different variables' values in characteristics of human face
- Represent values of the variables by their shape, size, placement and orientation.

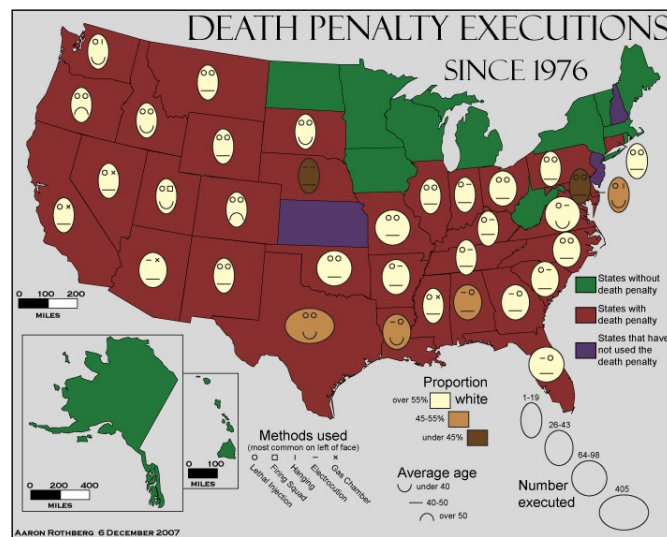


Jinwook Seo/John Stasko

Herman Chernoff (1973). "The Use of Faces to Represent Points in K-Dimensional Space Graphically". Journal of the American Statistical Association 68 (342): 361–368

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Chernoff Faces

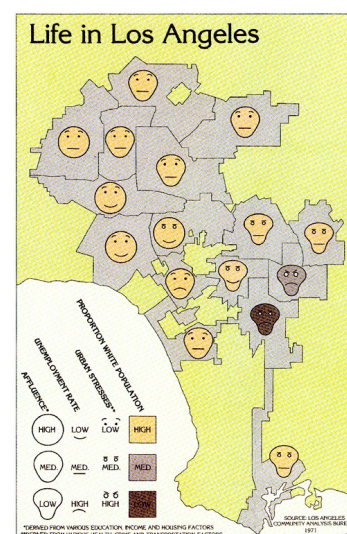


“visualized in the most
nightmarishly convoluted way
imaginable”

US states' death penalty executions since 1976

Chernoff Faces

- “Humans easily recognize faces and notice small changes without difficulty” ?
- eye size and eyebrow-slant

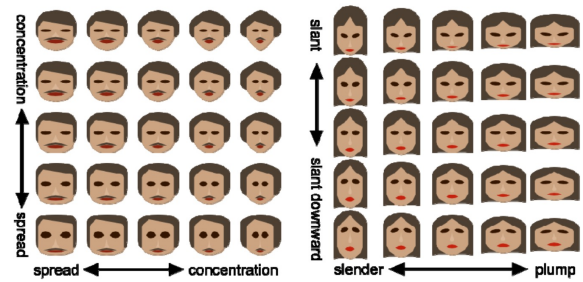


Chernoff Faces

- Combined with Apple Memoji & Generative model
 - Apple's Emoji character set includes seven faces for different ages and gender.
 - The Emoji characters also exist in five different skin colors.



Face variation using pixel-based images extracted from the Emoji character set of Apple's macOS 10.12.



Bisig, D. "2D Generative Faces for Evolutionary Social Simulation."

Multidimensional Data Visualizations

The Tables Lens

- Spreadsheet is certainly a multidimensional data presentation
- Idea: Make the text more **visual** and **symbolic**
- Just leverage basic **bar chart** idea

Encoding

- Change Quantitative values to bars

	A	B	C	D	E	F	G	H	I	J
1	ID	Name	HomeValue	%Income199	Poverty1999	PopDensity200	PopChange	Prcnt65+	Below18	PrcntFemal
2	1001	AL Autauga	94800	18518	10.9	43671	2.8	10.2	12494	51.4
3	1003	AL Baldwin	122500	20826	10.1	140415	3.8	15.5	34320	51
4	1005	AL Barbour	68600	13316	26.8	29038	-0.3	13.3	7383	48.4
5	1007	AL Bibb	74600	14105	20.6	20826	1.4	11.6	5286	48.4
6	1009	AL Blount	86800	16325	11.7	51024	2.4	12.9	12948	50.1
7	1011	AL Bullock	56600	10163	33.5	11714	-1.8	13.2	3058	47.6
8	1013	AL Butler	57700	15715	24.6	21399	-1.2			
9	1015	AL Calhoun	71600	17367	16.1	112249	-0.8			
10	1017	AL Chamber	58900	15147	17	36583	-0.3			
11	1019	AL Cherokee	76100	15543	15.6	23988	0.7			



<https://www.youtube.com/watch?v=ZDY9YCVv7z8>

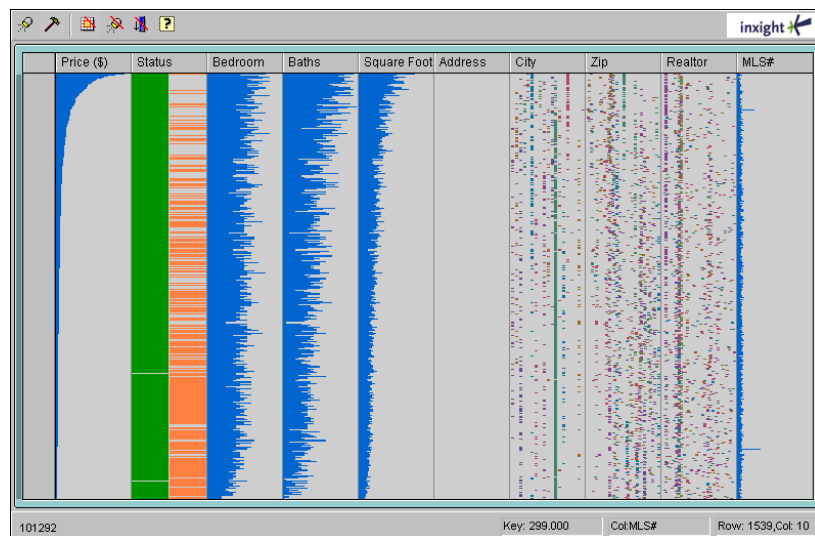
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What do you do for nominal data?

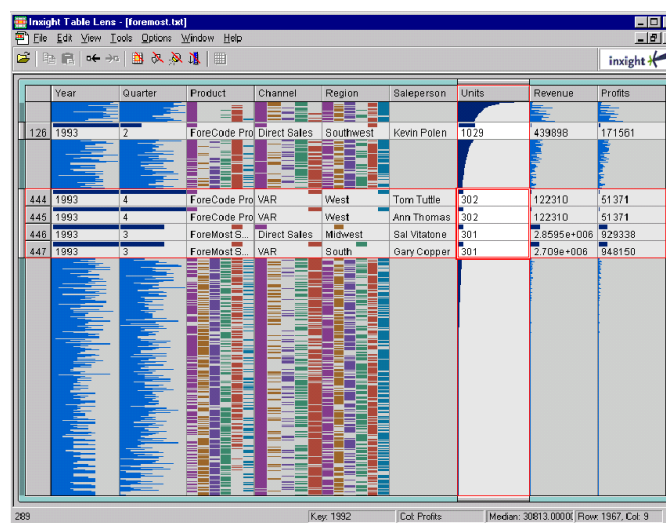
	A	B	C	D	E	F
1	cereal name	manufactu	type	calories	protein(g)	fat(g)
2	100% Bran	N	C	70	4	1
3	100%_Natural_Bran	Q	C	120	3	5
4	All-Bran	K	C	70	4	1
5	All-Bran_with_Extra_Fiber	K	C	50	4	0
6	Almond_Delight	R	C	110	2	2
7	Apple_Cinnamon_Cheerios	G	C	110	2	2
8	Apple_Jacks	K	C	110	2	0
9	Basic_4	G	C	130	3	2
10	Bran_CheX	R	C	90	2	1
11	Bran_Flakes	P	C	90	3	0



The Table Lens

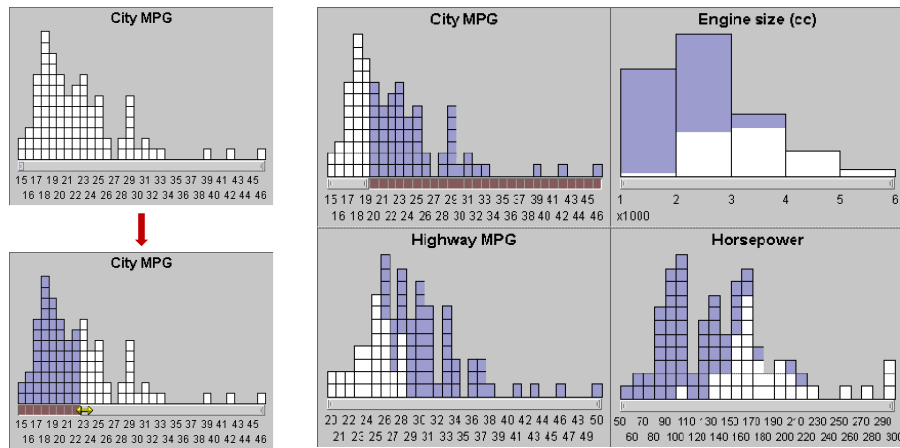


The Table Lens



Attribute Explorer

- Dynamic Query on Attributes

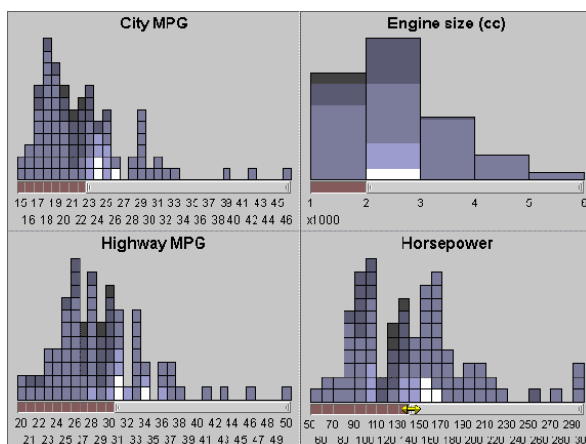


Tweedie, L., Spence, R., Williams, D.M.L., & Bhogal, R. (1994). The Attribute Explorer. *ACM, Conference Companion Proceedings CHI '94*, pp. 436-436.

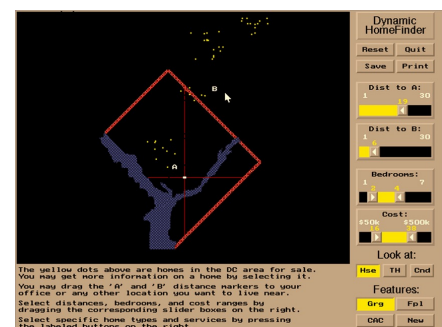
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Attribute Explorer

- color-coded with a progressively darker shade as they fail more constraints



Tweedie, L., Spence, R., Williams, D.M.L., & Bhogal, R. (1994). The Attribute Explorer. *ACM, Conference Companion Proceedings CHI '94*, pp. 436-436.



Christopher Williamson and Ben Shneiderman. 1992. The *Dynamic HomeFinder*: evaluating dynamic queries in a real-estate information exploration system. In *Proceedings of the 15th annual international ACM SIGIR conference on Research and development in information retrieval (SIGIR '92)*. Association for Computing Machinery, New York, NY, USA, 338-346. <https://doi.org/10.1145/133160.133216>

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Characteristics

- Multiple histogram views, one per attribute (like trellis)
- Each data case represented by a square
- Square is positioned relative to that case's value on that attribute
- Selecting a case in a view lights it up in others → **Brushing and Linking**
- Query sliders for narrowing → **Dynamic Query**
- Use **shading** to indicate **level of unsatisfied query match** → not widely accepted

Jinwook Seo/John Stasko

Video at <http://www.open-video.org/details.php?videoid=8162>

Information Visualization and Visual Analytics – Multidimensional data Visualization

Summary

- Features
 - Attribute histogram
 - Attribute relationship
 - Sensitivity information
 - Especially useful in “**zero-hits**” situations or when you are not familiar with the data at all

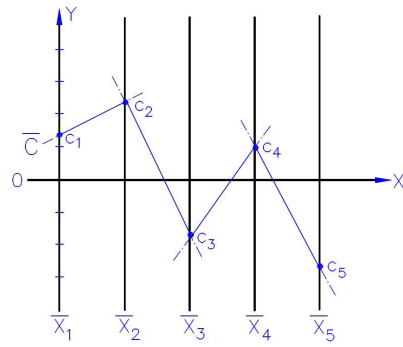
Jinwook Seo/John Stasko

Video at <http://www.open-video.org/details.php?videoid=8162>

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Parallel Coordinates Plot (PCP), ||-coords

- Each vertical bar represents a variable (and often has its own scale)
- Show how each variable is related to the others, especially adjacent ones



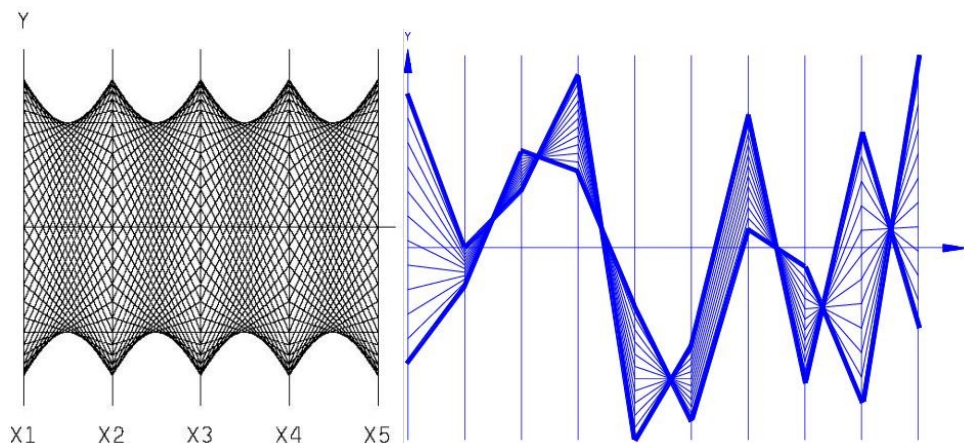
The point $C = (c_1, c_2, c_3, c_4, c_5)$



Al Inselberg, 1959

<http://www.math.tau.ac.il/~aisreal/>

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5-Dimensional Sphere

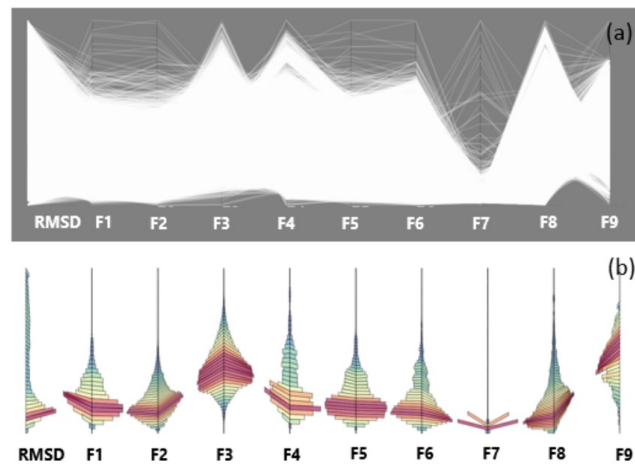
Line Interval In 10-D

<http://www.math.tau.ac.il/~aisreal/>

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Angular Histogram (AH)

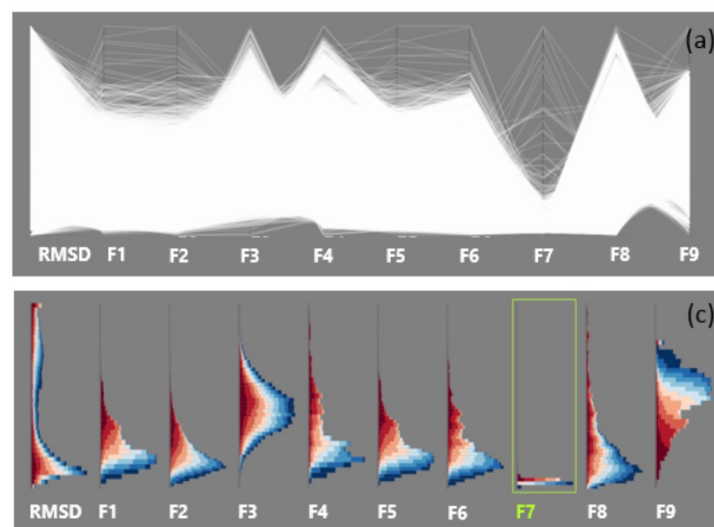
- Resolves overplotting problem of PCP



Bok, Jinwook, Bohyoung Kim, and Jinwook Seo. "Augmenting parallel coordinates plots with color-coded stacked histograms. TVCG 2021

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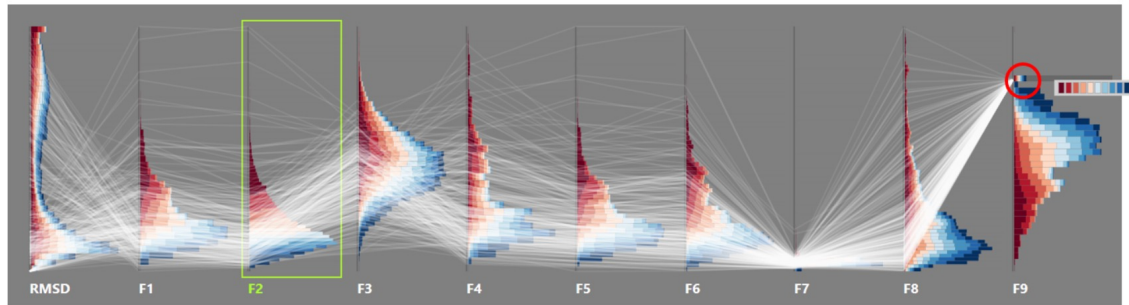
Parallel Histogram Plots (PHP)



Bok, Jinwook, Bohyoung Kim, and Jinwook Seo. "Augmenting parallel coordinates plots with color-coded stacked histograms. TVCG 2021

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Parallel Histogram Plots (PHP)



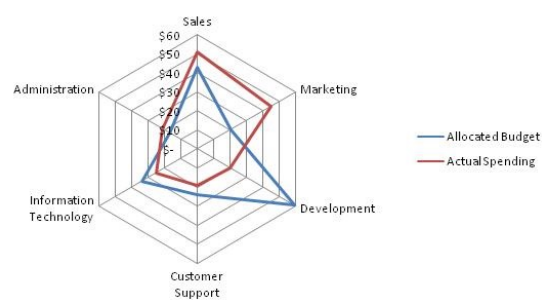
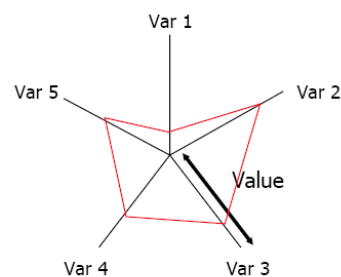
Bok, Jinwook, Bohyoung Kim, and Jinwook Seo. "Augmenting parallel coordinates plots with color-coded stacked histograms. TVCG 2021

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Multidimensional Data Visualizations

Star Plot (Radar Chart)

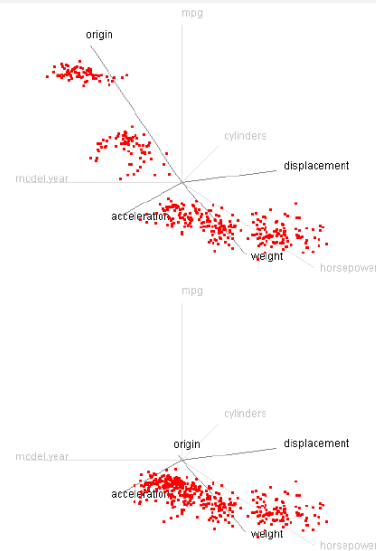
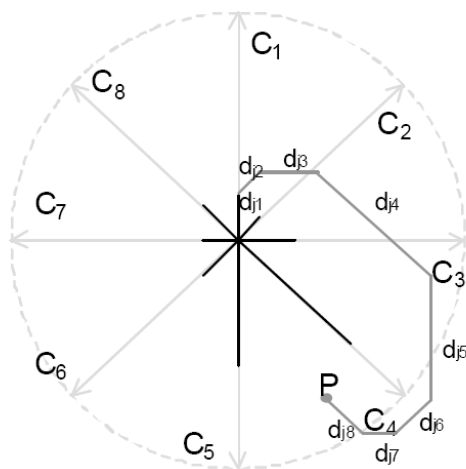
- Space out the n variables at equal angles around a circle
- Each “spoke” encodes a variable’s value
- Data point is now a “shape”



Jinwook Seo/John Stasko
http://en.wikipedia.org/wiki/File:Spider_Chart.jpg

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Star Coordinates



E. Kandogan, "Star Coordinates: A Multi-dimensional Visualization Technique with Uniform Treatment of Dimensions", InfoVis 2000

Jinwook Seo/John Stasko

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Star Coordinates - Summary

- Similar to star plot idiom
- Rather than represent point as polyline, just accumulate values along a *vector parallel to particular axis*
- Data case then becomes a **point**
- Find clusters and/or outliers
- Low dimensional *projection* methods

Jinwook Seo/John Stasko

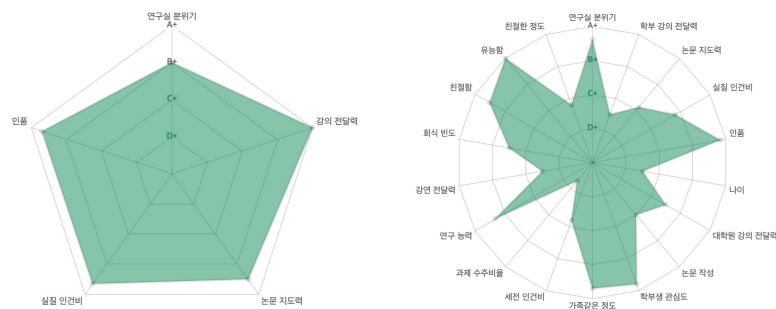
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Multidimensional vs. High-dimensional

- No specific boundary
 - Often used interchangeably
- **High-dimensional data** refers to data that cannot be effectively visualized using conventional multidimensional visualization techniques.
- **High-dimensional visualizations** refer to techniques specifically designed for representing such data.

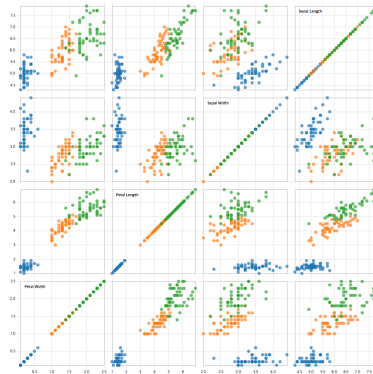
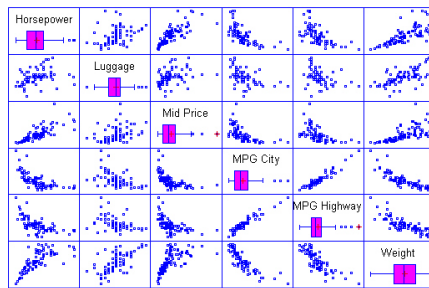
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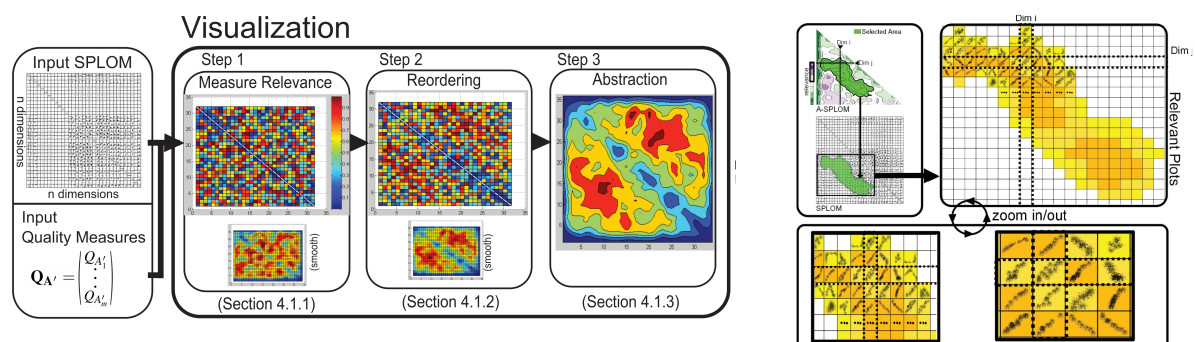
Scatterplot Matrix (SPLOM)

- Number of scatterplot required: $O(n^2)$
 - where n denotes the number of attributes
 - Makes SPLOM unscalable



Large Scatterplot Matrix

- High-dimensional version of scatterplot matrix
- Reorder & Abstract scatterplots to find interesting pattern



Multidimensional Projections (MDP)

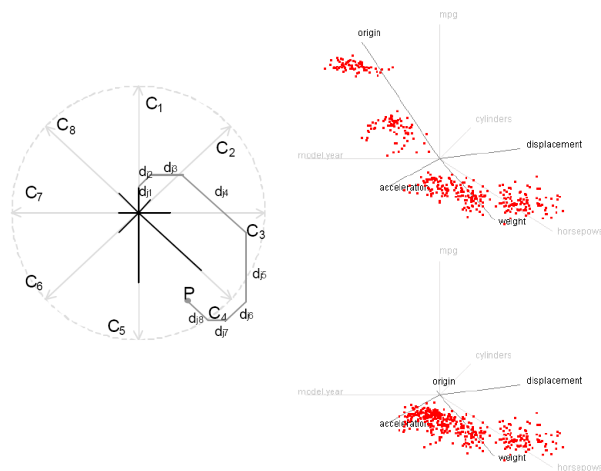
- Generate 2D projection of Multidimensional data
- Try to **preserve the characteristics of the original data** as much as they can
 - e.g., cluster structure, outlier, distances between points
- Each multidimensional data point is transformed to a 2D representation
 - Commonly represented as a point in a scatterplot



Multidimensional Projections

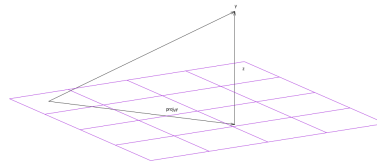
Star Coordinates

- Star Coordinates are not scalable for high-dimensional data!!



Linear Projections

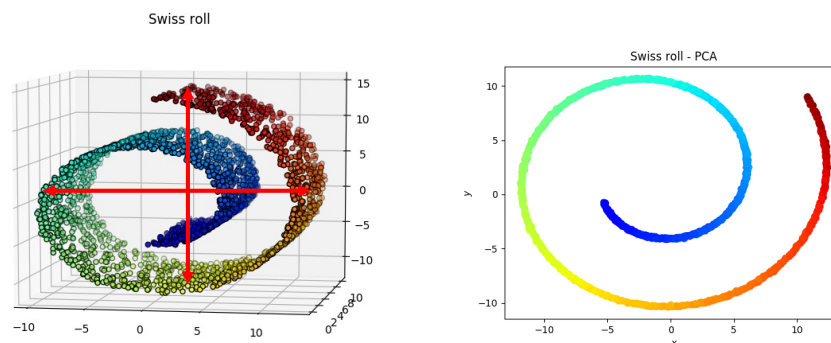
- Linear transformation of high-dimensional data



- Principal Component Analysis (PCA)
 - Find linear transformation which best preserves data's variation
- Multidimensional Scaling (MDS)
 - Find linear transformation that best preserves the pairwise distances between points

Linear Projections

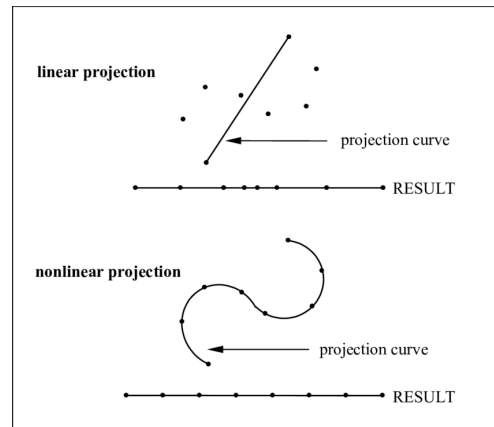
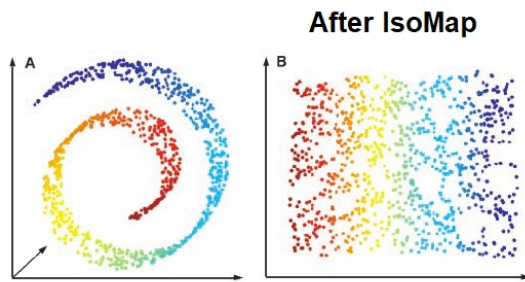
- Principal Component Analysis (PCA)
 - Find linear transformation which best preserves data's variation



Linear Projections

- Cannot “unfold” complex manifold in high-dimensional space

=> Nonlinear Projections arise!!



Kavzoglu, Taskin. (2001). An investigation of the design and use of feed-forward artificial neural networks in the classification of remotely sensed images.

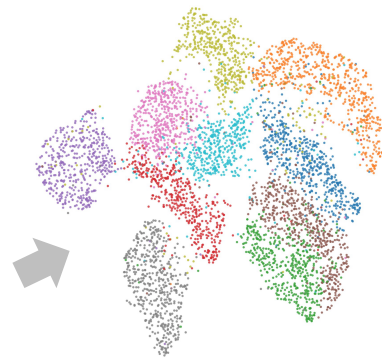
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Nonlinear Projections

- Anything that is not a linear projection
 - Features (e.g., clusters, manifolds) in high-dimensional space is nonlinear!!
 - Nonlinear projections aims to preserve those features
 - e.g., *t*-SNE, UMAP, Isomap, LLE
- Popular research topic in Machine learning / Big data / Data mining field
- Let's focus on how *visualization research field* treats projections

Nonlinear Projections

- t -SNE (t-distributed Stochastic Neighbor Embedding)
 - Machine-learning based nonlinear projections
 - Tries to match the data distribution of the original and projected space!!
 - Uses KL-divergence as loss function
 - Well-preserved pairwise similarity between points
 - One of the most widely used projection methods



Multidimensional Projections

 t -SNE

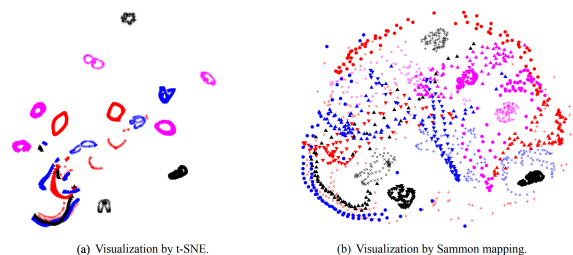
- Coil-20 dataset projections



(a) Example images of twenty categories

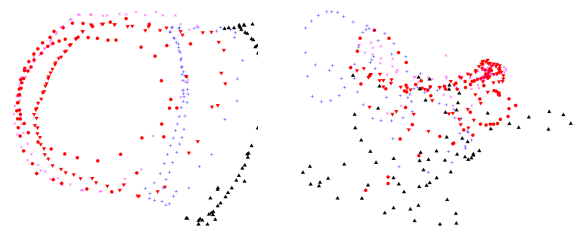


(b) Part example images of one category



(a) Visualization by t-SNE.

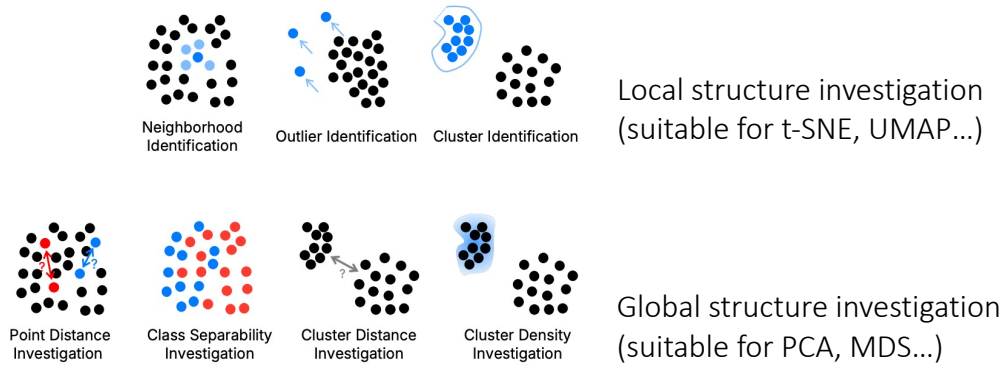
(b) Visualization by Sammon mapping.



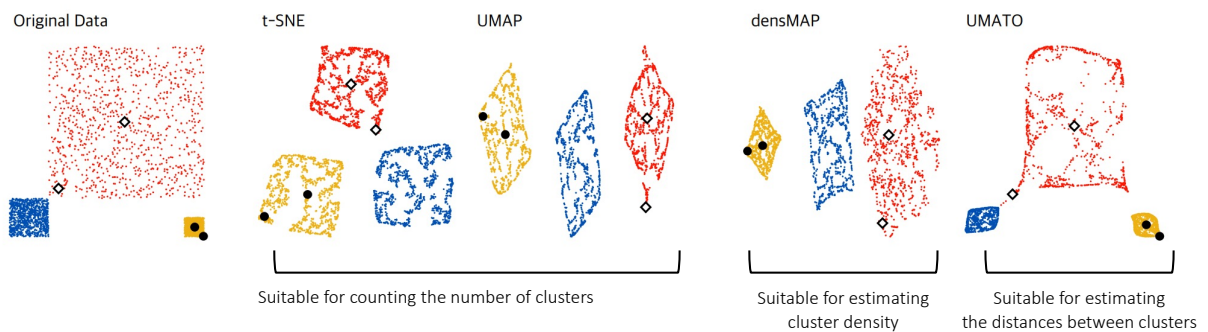
(c) Visualization by Isomap.

(d) Visualization by LLE.

Analytical Tasks in Multidimensional Projections

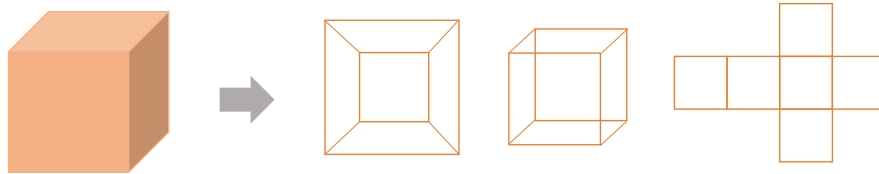


MDPs cannot show everything



Distortions

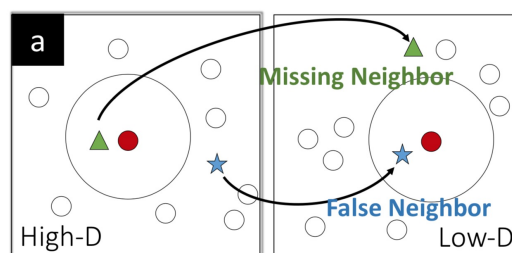
- Even a 3D cube cannot be exactly projected in 2D space!!



- The features of High-dimensional data are highly distorted in the projection
- *Multidimensional Projections suffer due to **distortions***

Distortions

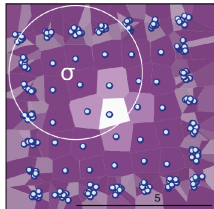
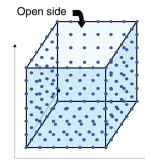
- Distortions in point-point relationships
 - **Missing Neighbors**
 - Neighbors in the original space are no more neighbors in the projection
 - **False Neighbors**
 - Neighbors that can be seen in the projection are actually not neighbors in the original space



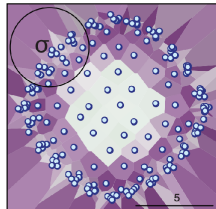
Distortions

• CheckViz

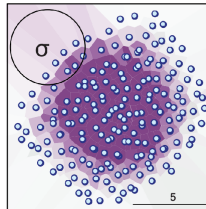
- Represents how much each point suffers from Missing / False Distortions
- Missing Neighbors: **Green**, False Neighbors: **Purple**



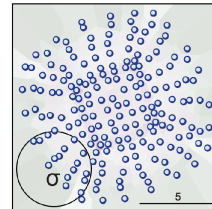
PCA



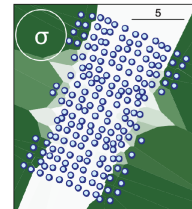
Isomap



NLM



DD-HDS



CCA

- Goal: Find the distortion patterns occurred in projections

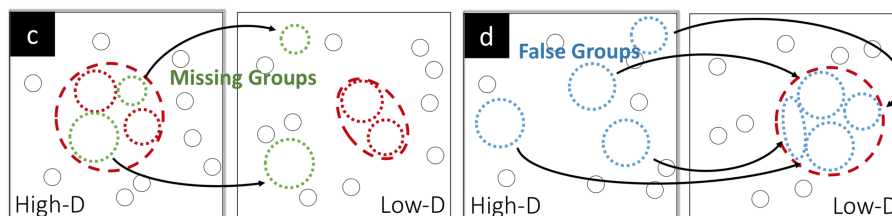
Lespinats, Sylvain, and Michaël Aupetit. "CheckViz: Sanity Check and Topological Clues for Linear and Non-Linear Mappings." *Computer Graphics Forum*.

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Distortions

• Missing Groups and False Groups distortions

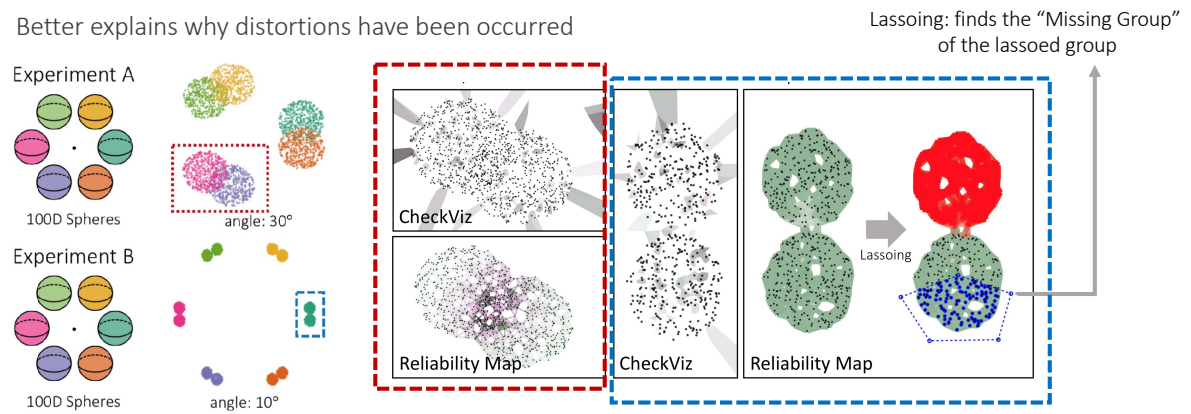
- Represents Missing / False concept in cluster level (cluster-cluster relationship)
- Better matches to visual analytics that focus on cluster analysis



Distortions

- Visualizing Missing / False Groups Distortions (Reliability map)

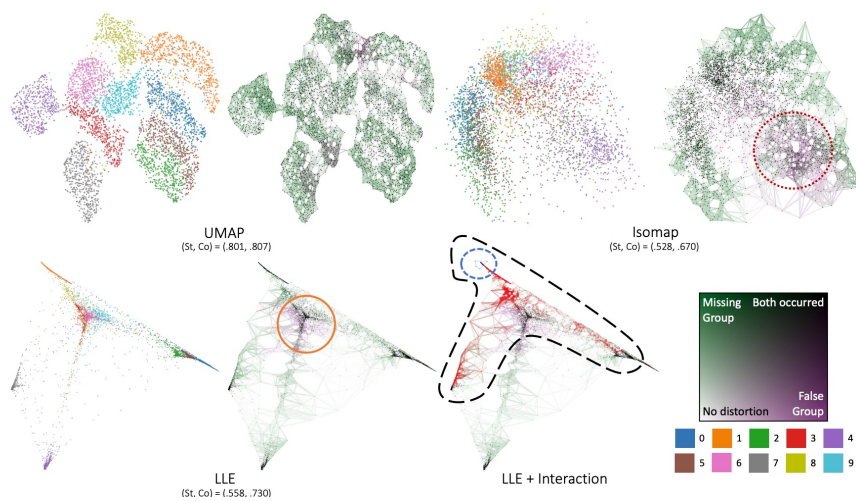
- Better represents the distortions pattern
- Better explains why distortions have been occurred



Hyeon Jeon et al. Measuring and Explaining the Inter-Cluster Reliability of Multidimensional Projections, TVCG 2021

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Distortions



Hyeon Jeon et al. Measuring and Explaining the Inter-Cluster Reliability of Multidimensional Projections, TVCG 2021

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Evaluating the Reliability of Dimensionality Reduction Embeddings

- ZADU and ZADUVis
- <https://github.com/hj-n/zadu>

Type	Measure	Ref.	provide pointwise distortions	dreval [39]	McInnes et al. [29]	Ingram et al. [15]	Jeon et al. [18]	Fujiwara et al. [10]	Espadoto et al. [9]	Colange et al. [6]	coranking [22]	pyclustering [33]	scikit-learn [35]	scipy [41]	Moor et al. [30]	Jeon et al. [19]	ZADU (Ours)
Local	Trustworthiness & Continuity	[40]	✓	△	△					○	○		△		○	○	○
	Mean Relative Rank Errors	[26]	✓														○
	Local Continuity Meta-Criteria	[4]	✓								○						○
	Neighborhood Hit	[34]	✓			○											○
	Neighbor Dissimilarity	[10]	✓					○									○
	Class-Aware Trustworthiness & Continuity	[6]	✓						○								○
Cluster-level	Procrustes Measure	[12]								○							○
	Steadiness & Cohesiveness	[18]	✓				○	○									○
	Distance Consistency	[37]															○
	Internal Clustering Validation Measures	[21]										○	○				○
Global	Clustering + External Clustering Validation Measures	[42]											○	○			○
	Stress	[23, 24]															○
	Kullback-Leibler Divergence	[13]							○							○	○
	Distance-to-Measure	[3]														○	○
	Topographic Product	[1]														○	○
	Pearson's correlation coefficient r	[11]													○		○
	Spearman's rank correlation coefficient ρ	[36]													○		○

Evaluating the Reliability of Dimensionality Reduction Embeddings

- ZADU and ZADUVis

```

from zadu import zadu
from zaduvis import zaduvis
import matplotlib.pyplot as plt
from sklearn.manifold import TSNE
from sklearn.datasets import fetch_openml

hd = fetch_openml('mnist_784', version=1, cache=True).data.to_numpy()[1::7]
ld = TSNE().fit_transform(hd)

## Computing local pointwise distortions
spec = {
    "id": "tnc",
    "params": {"k": 25}
}, {
    "id": "snc",
    "params": {"k": 50}
}

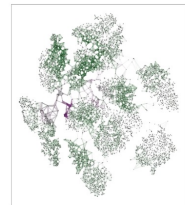
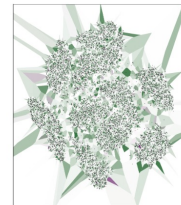
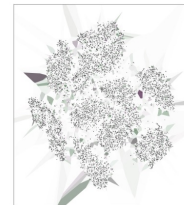
zadu_obj = zadu.ZADU(spec, hd, return_local=True)
scores, local_list = zadu_obj.measure(ld)

tnc_local = local_list[0]
snc_local = local_list[1]

local_trustworthiness = tnc_local["local_trustworthiness"]
local_continuity = tnc_local["local_continuity"]
local_steadiness = snc_local["local_steadiness"]
local_cohesiveness = snc_local["local_cohesiveness"]

fig, ax = plt.subplots(1, 4, figsize=(50, 12.5))
zaduvis.checkviz(ld, local_trustworthiness, local_continuity, ax=ax[0])
zaduvis.reliability_map(ld, local_trustworthiness, local_continuity, k=10, ax=ax[1])
zaduvis.checkviz(ld, local_steadiness, local_cohesiveness, ax=ax[2])
zaduvis.reliability_map(ld, local_steadiness, local_cohesiveness, k=10, ax=ax[3])

```



- Questions?